INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

CIP-B101 Lab 1B

Basic Computer Skills for Digital Forensics

PC System Fundamentals, Disk Sectors and Partition Analysis

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Instructor _____

Practical environment Windows 11 Pro on a Dell Latitude 7430

Introduction

This practical examines a Windows computer using read-only tools.
It records hardware, operating system, boot mode, and storage details.
It identifies evidence at the user, application, OS, disk, and process layers.
It compares partition information across three Windows tools.
It uses logical sector size to verify the boundaries and sizes of two partitions.
It explains how the findings support digital forensic examination.

Tools used

System Information (msinfo32), Command Prompt (systeminfo), Windows PowerShell (Get-Disk, Get-Volume, Get-Process and Get-Partition), Disk Management, and DiskPart listing commands. Partition calculations were checked in PowerShell.

Course material *Computer System Introduction* and the CIP-B101 Lab 1B practical manual.

SYSTEM IDENTIFICATION

Windows system profile

Item	Observed value
Operating system	Microsoft Windows 11 Pro, version 10.0.26200, build 26200
System	Dell Latitude 7430, x64-based PC
Processor	12th Gen Intel Core i7-1265U, 10 cores and 12 logical processors
Memory	16.0 GB installed; Windows System Information reports 15.4 GB total physical memory
Firmware and boot	Dell firmware 1.42.0; UEFI; Secure Boot On
Time zone	East Africa Standard Time (UTC+03:00)



Figure 1a. Full Windows System Information screenshot, including UEFI and Secure Boot (evidence items 1 and 8).

System information and RAM

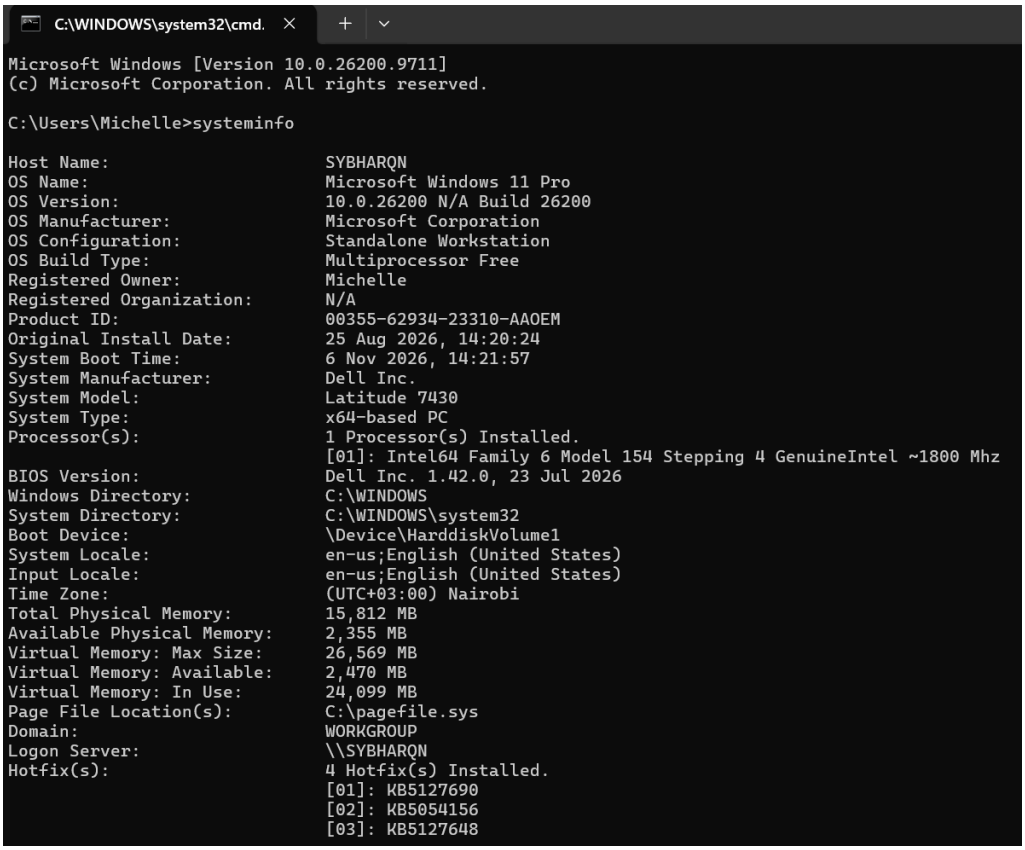


Figure 1b. Command Prompt systeminfo output.

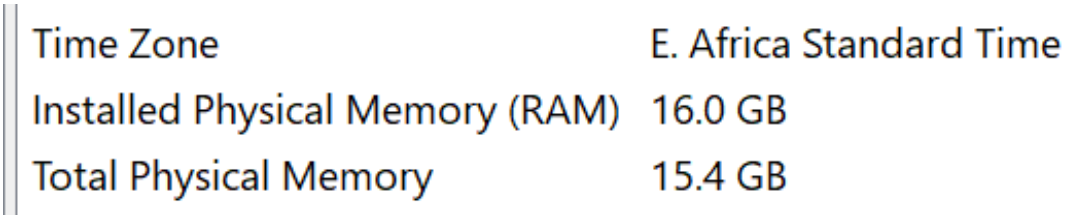


Figure 1c. System Information showing 16.0 GB of installed RAM.

Evidence mapping and live processes

Layer or source	Example evidence	Where it may be found	Forensic value
User	Recent file access	Recent Items shortcuts	Helps associate a user account with file access.
Application	Browsing history	Browser profile database	Helps reconstruct visited pages and activity times.
Operating system	Logon event	Windows Security event log, if auditing records it	Helps identify the account and time of a logon.
Disk	File metadata or deleted file remnants	NTFS MFT and unallocated space	Helps reconstruct file history or locate recoverable content.
Memory and processes	Running process name and PID	Live process list or memory capture	Helps identify what was running when collected.

One action can leave traces across layers. Opening a file may create an application recent-file entry and an operating system shortcut. Comparing those records can help establish the sequence of activity.

Process observation

Process	PID	Likely purpose
cmd	24256	Windows Command Prompt.
conhost	9160	Windows Console Host for console applications.
audiodg	22148	Windows audio engine process for audio processing.

A program is stored code. A process is a running instance with its own resources and PID. A thread is a unit of execution within a process. The snapshot shows several conhost instances with different PIDs. In live triage, a PID can help correlate a running process with other observations from the same session. A process name alone cannot establish whether it is legitimate.

```
PS C:\Users\Michelle> Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet

Name                     Id CPU       WorkingSet
----                     -- ---       ----------
AggregatorHost         3820             13967360
AppActions            15524 3.140625    27979776
ApplicationFrameHost  27460 1.796875    35778560
AppVShNotify          10164 0.078125    10137600
audiodg               22148 941.3125   149155840
backgroundTaskHost     8700 0.453125    59723776
bcmHostStorageService  5036             12185600
bcmUshUpgradeService   3520              8605696
cmd                   24256 0.015625     8835072
conhost                9160              7909376
conhost                9240              6447104
conhost               11608 0.015625     5840896
conhost               22352 0.03125     10997760
conhost               22920 0.015625    10997760
crashpad_handler        488 0.046875     7938048
CrossDeviceResume      5976 1.671875    13938688
csrss                  1124              4747264
csrss                  1220              7942144
ctfmon                11164 55.109375   23838720
dasHost                4972              7114752
```

Figure 4. Running process names, PIDs, CPU time, and working set.

STORAGE EVIDENCE

Disk capacity and logical volumes

Disk 0 is a WDC PC SN730 SDBQNTY-512G-1001. PowerShell reports 512,110,190,592 bytes, approximately 512.11 GB or 476.94 GiB. The disk uses GPT and reports 512-byte logical sectors and 4,096-byte physical sectors. Logical sectors are the units used for the partition calculations.

```
PS C:\WINDOWS\system32> Get-Disk | Format-List Number,FriendlyName,Size,PartitionStyle,LogicalSectorSize,PhysicalSectorSize


Number             : 0
FriendlyName       : WDC PC SN730 SDBQNTY-512G-1001
Size               : 512110190592
PartitionStyle     : GPT
LogicalSectorSize  : 512
PhysicalSectorSize : 4096
```

Figure 2. Disk model, exact capacity, GPT partition style, and sector sizes.

Volume	File system	Size (bytes)	Free space (bytes)
C:	NTFS	345,572,900,864	147,502,379,008
D: (New Volume)	NTFS	164,572,950,528	125,100,609,536
No drive letter shown	NTFS	962,588,672	120,373,248

```
PS C:\Users\Michelle> Get-Volume | Format-Table DriveLetter,FileSystemLabel,FileSystem,Size,SizeRemaining -AutoSize

DriveLetter FileSystemLabel FileSystem         Size SizeRemaining
----------- --------------- ----------         ---- -------------
          D New Volume      NTFS       164572950528  125100609536
          C                 NTFS       345572900864  147502379008
                            NTFS          962588672     120373248
```

Figure 3. Get-Volume output for logical volumes and file-system identification (evidence items 3 and 9).

C: and D: correspond to partitions 3 and 5 in Get-Partition. Both use NTFS. An examiner can use NTFS metadata such as the Master File Table for names and timestamps, and consult the USN change journal where available to help reconstruct changes.

Disk 0 partition layout

Disk Management shows the graphical layout, and Get-Partition shows the exact byte offsets and sizes. DiskPart provides a second listing of the partition numbers and types. The inspection used only read-only queries and DiskPart listing commands.

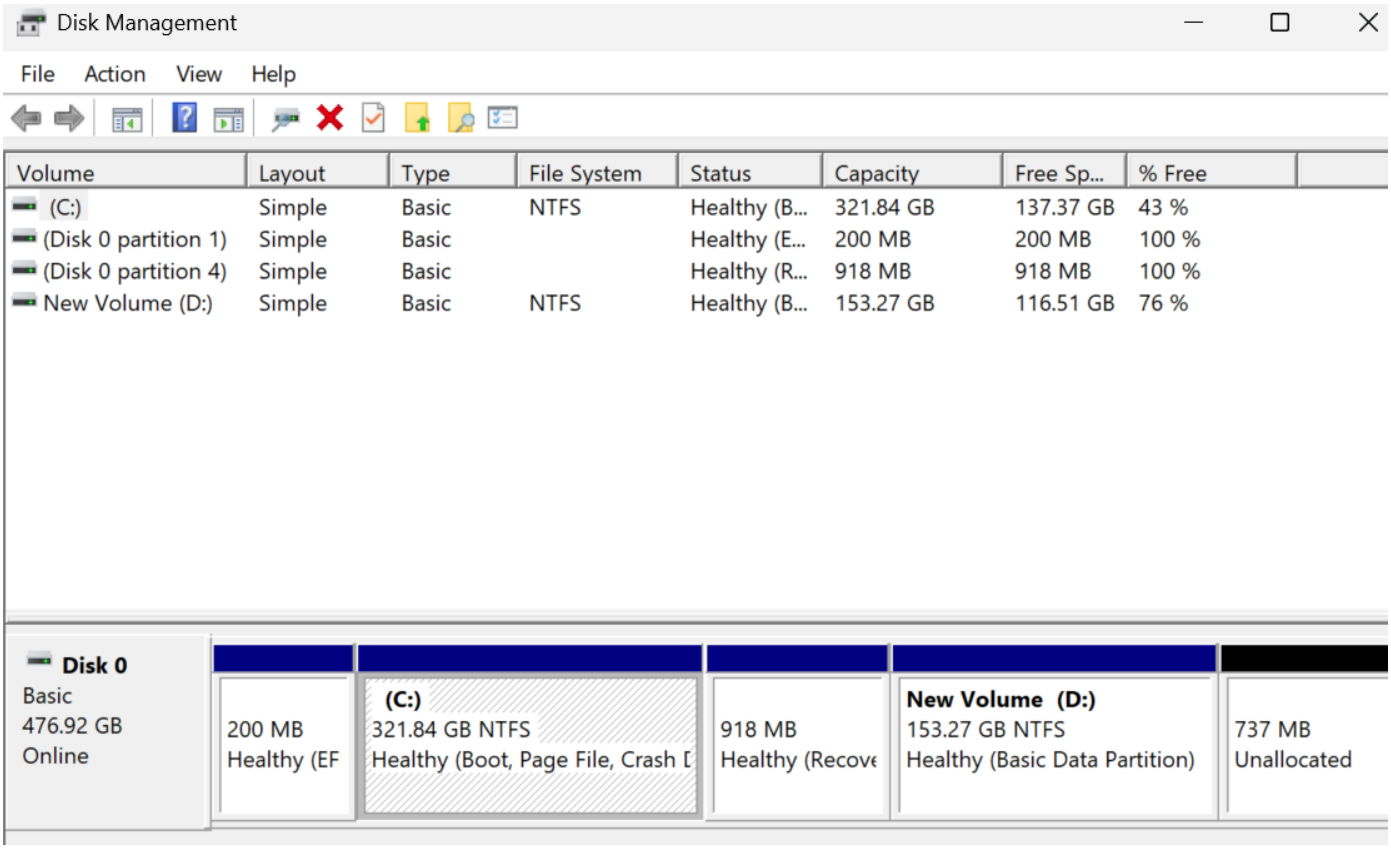


Figure 5a. Disk Management showing Disk 0 and its visible partitions.

```
PS C:\Users\Michelle> Get-Partition | Sort-Object DiskNumber,PartitionNumber |
>>     Format-Table DiskNumber,PartitionNumber,DriveLetter,Type,Offset,Size -AutoSize

DiskNumber PartitionNumber DriveLetter Type           Offset         Size
---------- --------------- ----------- ----           ------         ----
         0               1             System        1048576    209715200
         0               2             Reserved    210763776     16777216
         0               3           C Basic       227540992 345572901376
         0               4             Recovery 345801490432    962592768
         0               5           D Basic    346764083200 164572954624
```

Figure 5b. Get-Partition output showing exact byte offsets and sizes.

DiskPart and slide calculation

```
C:\Windows\System32>diskpart

Microsoft DiskPart version 10.0.26100.1150

Copyright (C) Microsoft Corporation.
On computer: SYBHARQN

DISKPART> list disk

  Disk ###  Status         Size     Free     Dyn  Gpt
  --------  -------------  -------  -------  ---  ---
  Disk 0    Online          476 GB   737 MB        *

DISKPART> select disk 0

Disk 0 is now the selected disk.

DISKPART> list partition

  Partition ###  Type              Size     Offset
  -------------  ----------------  -------  -------
  Partition 1    System             200 MB  1024 KB
  Partition 2    Reserved            16 MB   201 MB
  Partition 3    Primary            321 GB   217 MB
  Partition 4    Recovery           918 MB   322 GB
  Partition 5    Primary            153 GB   322 GB

DISKPART> exit

Leaving DiskPart...
```

Figure 5c. DiskPart listing for Disk 0, including the reserved Partition 2.

Part C1. Disk capacity from the slide example

Step	Calculation	Result
Total sectors	400 × 12 × 17,000	81,600,000 sectors
Total bytes	81,600,000 × 512	41,779,200,000 bytes
Convert to KiB	41,779,200,000 ÷ 1,024	40,800,000 KiB
Convert to MiB	40,800,000 ÷ 1,024	39,843.75 MiB
Convert to GiB	39,843.75 ÷ 1,024	38.91 GiB (rounded)

PARTITION CALCULATIONS

Disk 0 partitions 1 and 2

Inputs are the exact offsets and sizes in Figure 5b and the 512-byte logical sector size in Figure 2. The final sector is inclusive. GiB uses 1,073,741,824 bytes per unit.

Step	**Partition 1 · EFI System**	**Partition 2 · Microsoft Reserved**
Start sector	1,048,576 ÷ 512 = 2,048	210,763,776 ÷ 512 = 411,648
Sector count	209,715,200 ÷ 512 = 409,600	16,777,216 ÷ 512 = 32,768
Last sector	2,048 + 409,600 − 1 = 411,647	411,648 + 32,768 − 1 = 444,415
Size in bytes	409,600 × 512 = 209,715,200	32,768 × 512 = 16,777,216
Size in MiB	209,715,200 ÷ 1,048,576 = 200	16,777,216 ÷ 1,048,576 = 16
Size in GiB	209,715,200 ÷ 1,073,741,824 = 0.1953125	16,777,216 ÷ 1,073,741,824 = 0.015625

Disk Management and DiskPart display Partition 1 as 200 MB. DiskPart displays Partition 2 as 16 MB. These are the same numerical values as the calculated MiB sizes. The reserved partition is absent from the captured Disk Management view. Neither partition has a drive letter or a file system identified in the captured volume output.

PowerShell verification

```
PS C:\Users\Michelle> 'Disk 0, Partition 1 - EFI System'
Disk 0, Partition 1 - EFI System
PS C:\Users\Michelle> "Starting sector: $(1048576 / 512)"
Starting sector: 2048
PS C:\Users\Michelle> "Sector count: $(209715200 / 512)"
Sector count: 409600
PS C:\Users\Michelle> "Last sector: $(2048 + 409600 - 1)"
Last sector: 411647
PS C:\Users\Michelle> "Verified bytes: $(409600 * 512)"
Verified bytes: 209715200
PS C:\Users\Michelle> "Size in MiB: $(209715200 / 1MB)"
Size in MiB: 200
PS C:\Users\Michelle> "Size in GiB: $(209715200 / 1GB)"
Size in GiB: 0.1953125
PS C:\Users\Michelle>
```

Figure 6. Partition 1 calculation and PowerShell results.

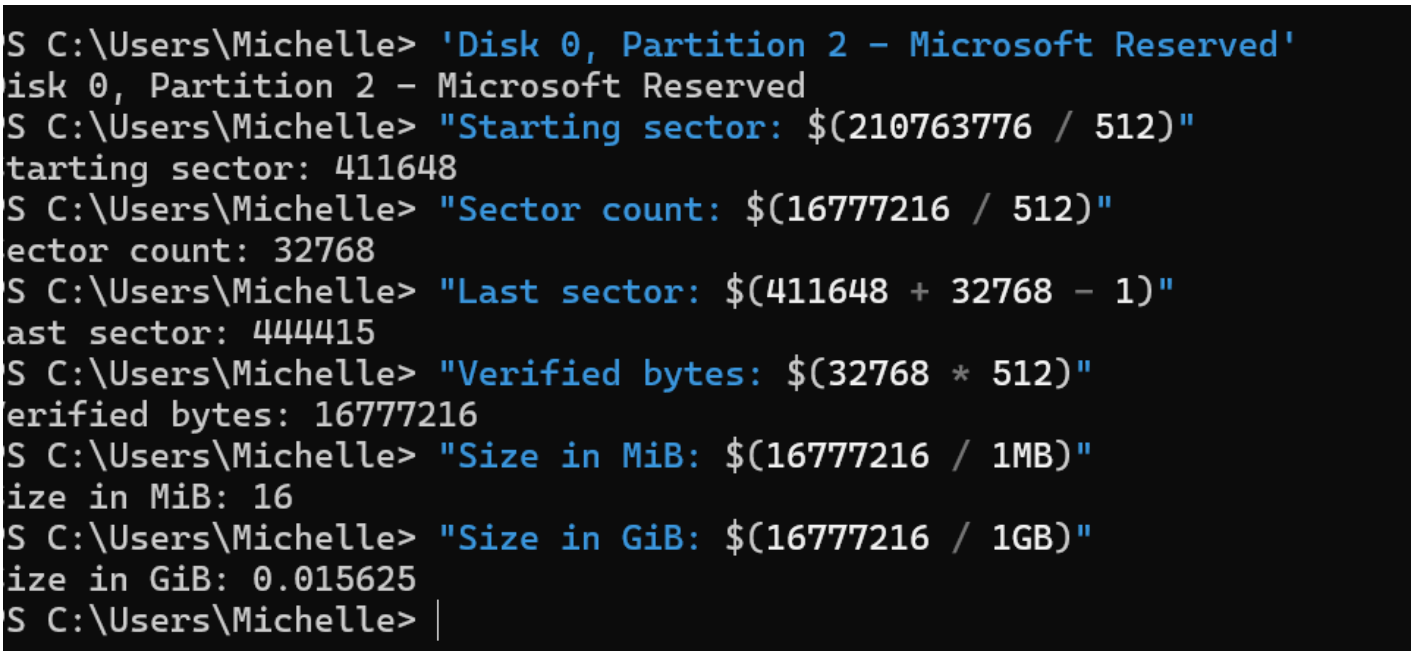


Figure 7. Partition 2 calculation and PowerShell results.

Boot sequence and MBR versus GPT

System Information reports **UEFI** and **Secure Boot On** (Figure 1a). Get-Disk reports **GPT** (Figure 2), and DiskPart identifies Partition 1 as System (Figure 5c).

Stage	What happens
1. Power-on and POST	Firmware initializes and checks essential hardware.
2. Firmware settings	Firmware applies boot order and security settings. Legacy systems commonly refer to CMOS settings. UEFI uses nonvolatile firmware variables for boot entries.
3. Locate boot code	Legacy BIOS loads boot code from the disk MBR. On this UEFI/GPT system, the firmware boot entry points to an EFI application on the EFI System Partition.
4. Boot manager	Windows Boot Manager reads boot configuration and starts the selected Windows loader. Secure Boot checks signatures as part of the UEFI boot chain.
5. Kernel and OS control	The Windows loader loads the kernel and essential boot drivers. The kernel takes control and starts the operating system.

Feature	MBR	GPT
Partition metadata	Partition entries and legacy boot code are stored in sector 0.	Primary header and entries near the start, with backup copies near the end.
Partition count	Four primary entries. An extended partition can contain logical partitions.	Windows typically supports up to 128 partitions on a GPT disk.
Capacity	About 2 TiB with 512-byte logical sectors.	Uses 64-bit sector addressing and supports much larger disks.
Forensic relevance	Damage to sector 0 can obscure the partition layout.	Backup metadata and checksums can help validate or recover the partition layout.

SUMMARY

Conclusion and evidence checklist

The Windows computer booted in UEFI mode with Secure Boot enabled. Its internal disk used GPT. Partition 1 was the EFI System Partition at sectors 2,048 to 411,647 and measured 200 MiB. Partition 2 was Microsoft Reserved at sectors 411,648 to 444,415 and measured 16 MiB. Both results agreed with DiskPart. The C: and D: volumes were NTFS.

Using exact byte offsets and the reported logical sector size made the calculation reproducible. The reserved partition was visible in DiskPart and PowerShell even though it did not appear in the captured Disk Management layout. Checking more than one view helped account for disk areas that a volume listing alone would miss.

Required evidence	**Report reference**
1. System Information or systeminfo	Figures 1a–1c
2. Disk model and storage capacity	Figure 2
3. Logical volume and file-system information	Figure 3
4. Running processes and PID information	Figure 4
5. Disk and partition table inspection	Figures 5a–5c
6. Partition 1 calculation and evidence	Calculation table and Figure 6
7. Partition 2 calculation and evidence	Calculation table and Figure 7
8. BIOS/UEFI boot mode evidence	Figure 1a and boot section
9. File-system identification evidence	Figure 3 and storage section
10. Final summary and conclusion	This page